

Background: Navigation Policies

The navigation behavior of the agent is governed by a combination of local obstacle avoidance and global goal-seeking.

Egocentric Policy

The egocentric policy reacts to the local occupancy of neighboring cells, penalizing blocked directions with a repulsion strength $\alpha > 0$:

$$\pi_E(a \mid \mathbf{s}) = \frac{\exp[-\alpha O_a]}{\sum_{a'} \exp[-\alpha O_{a'}]}, \quad (1)$$

where $O_a \in \{0, 1\}$ is the occupancy of the cell reached by action a .

Geocentric Policy

The geocentric policy biases the agent toward the goal $\mathbf{g}$. It is a softmax over the alignment scores $\phi_a = \hat{\mathbf{e}}_a \cdot \hat{\mathbf{u}}$, where $\hat{\mathbf{u}} = (\mathbf{g} - \mathbf{s})/\|\mathbf{g} - \mathbf{s}\|$, with inverse temperature $\beta > 0$:

$$\pi_G(a \mid \mathbf{s}) = \frac{\exp[\beta \phi_a]}{\sum_{a'} \exp[\beta \phi_{a'}]}. \quad (2)$$

Mixture Policy

The combined stochastic policy is a convex mixture of the geocentric and egocentric policies, controlled by the mixing parameter $\gamma \in [0, 1]$:

$$\pi(a \mid \mathbf{s}) = \gamma \pi_G(a \mid \mathbf{s}) + (1 - \gamma) \pi_E(a \mid \mathbf{s}). \quad (3)$$

Discrete Master Equation

Let $P(\mathbf{r}, t)$ be the probability of finding the agent at position $\mathbf{r}$ at time t . The discrete-time evolution is given by the Chapman–Kolmogorov master equation:

$$P(\mathbf{r}, t + \Delta t) - P(\mathbf{r}, t) = \sum_{\mathbf{a}} [\pi(\mathbf{a} \mid \mathbf{r} - \mathbf{a}) P(\mathbf{r} - \mathbf{a}, t) - \pi(\mathbf{a} \mid \mathbf{r}) P(\mathbf{r}, t)]. \quad (4)$$

Taking the continuum limit of this master equation via the Kramers–Moyal expansion (truncating at the second moment) yields the Fokker–Planck Equation (FPE).

Linearized Fokker–Planck Equation

Under a small angle approximation, the resulting linearized Fokker–Planck equation is:

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x} [V_x(\mathbf{r}) P] - \frac{\partial}{\partial y} [V_y(\mathbf{r}) P] + D_{\parallel} \frac{\partial^2 P}{\partial x^2} + D_{\perp} \frac{\partial^2 P}{\partial y^2}. \quad (5)$$

Coefficients of the Fokker–Planck Equation

Correspondence Between FPE and SDE

Every Fokker–Planck equation of the form

$$\frac{\partial P}{\partial t} = - \sum_i \frac{\partial}{\partial x_i} [\mu_i(\mathbf{x})P] + \frac{1}{2} \sum_{i,j} \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(\mathbf{x})P] \quad (12)$$

corresponds to a general Itô Stochastic Differential Equation (SDE):

$$d\mathbf{X}_t = \boldsymbol{\mu}(\mathbf{X}_t)dt + \boldsymbol{\sigma}(\mathbf{X}_t)d\mathbf{W}_t, \quad (13)$$

where $\mathbf{W}_t$ is a vector of independent Wiener processes. The diffusion tensor $\mathbf{D}(\mathbf{x})$ is related to the noise coefficient matrix $\boldsymbol{\sigma}(\mathbf{x})$ by:

$$\mathbf{D}(\mathbf{x}) = \boldsymbol{\sigma}(\mathbf{x})\boldsymbol{\sigma}(\mathbf{x})^\top. \quad (14)$$

Vector Form of the SDE for the 2D System

For our two-component system with state vector $\mathbf{r}_t = (x_t, y_t)$, the drift vector is $\mathbf{V}(\mathbf{r}) = (V_x(\mathbf{r}), V_y(\mathbf{r}))$. The diffusion matrix is diagonal, $\mathbf{D} = \text{diag}(2D_\parallel, 2D_\perp)$. Thus, we can define the noise coefficient matrix $\mathbf{B} = \text{diag}(\sqrt{2D_\parallel}, \sqrt{2D_\perp})$ such that $\mathbf{B}\mathbf{B}^\top = \mathbf{D}$.

The corresponding SDE in vector form is:

$$d\mathbf{r}_t = \mathbf{V}(\mathbf{r}_t)dt + \mathbf{B}d\mathbf{W}_t, \quad (15)$$

or explicitly,

$$\begin{pmatrix} dx_t \\ dy_t \end{pmatrix} = \begin{pmatrix} V_x(\mathbf{r}_t) \\ V_y(\mathbf{r}_t) \end{pmatrix} dt + \begin{pmatrix} \sqrt{2D_\parallel} & 0 \\ 0 & \sqrt{2D_\perp} \end{pmatrix} \begin{pmatrix} dW_{x,t} \\ dW_{y,t} \end{pmatrix}. \quad (16)$$

Objective

The objective is to recover a reward functional that the agent may be optimizing, and the corresponding value function, in both the discrete and continuum cases.

In the discrete case, the aim is to recover a discrete reward functional and the corresponding discrete value function. In the continuum case, the aim is to recover the continuum stochastic control reward, or cost, and the corresponding value function associated with the SDE.

Assumptions Required Before Proceeding

We write the vector-form SDE corresponding to the linearized Fokker–Planck equation as:

$$d\mathbf{r}_t = \mathbf{V}(\mathbf{r}_t) dt + \mathbf{B} d\mathbf{W}_t, \quad (17)$$

where position $\mathbf{r}_t$, velocity $\mathbf{V}(\mathbf{r})$, and the noise matrix $\mathbf{B}$ are defined as:

$$\mathbf{r}_t = \begin{pmatrix} x_t \\ y_t \end{pmatrix}, \quad \mathbf{V}(\mathbf{r}) = \begin{pmatrix} V_x(\mathbf{r}) \\ V_y(\mathbf{r}) \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \sqrt{2D_{\parallel}} & 0 \\ 0 & \sqrt{2D_{\perp}} \end{pmatrix}. \quad (18)$$

To treat this system as a stochastic optimal control problem, we state two core modeling assumptions:

1. **Control Acts Only on Drift:** The active control force $\mathbf{u}_t$ directly shifts the system’s velocity drift, leaving the diffusion noise matrix $\mathbf{B}$ unchanged.
2. **Fixed Exogenous Diffusion:** Although anisotropic diffusion ($D_{\parallel} \neq D_{\perp}$) originally arose from the geocentric policy in the moment derivation, we treat $\mathbf{B}$ as a fixed environmental diffusion structure rather than a control-dependent diffusion.

Fixing $\mathbf{B}$ ensures that the diffusion tensor remains identical under both the reference and controlled measures, satisfying the fundamental requirement for Girsanov’s theorem.

Control Cost from Girsanov’s Theorem

Girsanov’s theorem defines a change of probability measure that shifts the drift of a stochastic process while preserving its noise structure. Consider an uncontrolled reference measure $\mathbb{P}^0$ with passive drift $\mathbf{f}(\mathbf{r}_t)$:

$$d\mathbf{r}_t = \mathbf{f}(\mathbf{r}_t) dt + \mathbf{B} d\mathbf{W}_t^0, \quad (19)$$

and a controlled measure $\mathbb{P}^{\mathbf{u}}$ with modified drift $\mathbf{f}(\mathbf{r}_t) + \mathbf{u}_t$:

$$d\mathbf{r}_t = [\mathbf{f}(\mathbf{r}_t) + \mathbf{u}_t] dt + \mathbf{B} d\mathbf{W}_t^{\mathbf{u}}. \quad (20)$$

The effective shift in the standard Brownian noise vector is given by the Girsanov kernel $\mathbf{k}_t = \mathbf{B}^{-1}\mathbf{u}_t$. The Radon–Nikodym derivative relating the two path measures is:

$$\frac{d\mathbb{P}^{\mathbf{u}}}{d\mathbb{P}^0} = \exp \left(\int_0^T \mathbf{k}_t^{\top} d\mathbf{W}_t^0 - \frac{1}{2} \int_0^T \|\mathbf{k}_t\|^2 dt \right). \quad (21)$$

The Kullback–Leibler (KL) divergence between the controlled path measure $\mathbb{P}^{\mathbf{u}}$ and the reference measure $\mathbb{P}^0$ measures the statistical effort required to maintain the control $\mathbf{u}_t$:

$$D_{\text{KL}}(\mathbb{P}^{\mathbf{u}} \parallel \mathbb{P}^0) = \mathbb{E}_{\mathbb{P}^{\mathbf{u}}} \left[\frac{1}{2} \int_0^T \|\mathbf{k}_t\|^2 dt \right] = \mathbb{E}_{\mathbb{P}^{\mathbf{u}}} \left[\frac{1}{2} \int_0^T \mathbf{u}_t^{\top} (\mathbf{B}\mathbf{B}^{\top})^{-1} \mathbf{u}_t dt \right]. \quad (22)$$

The integrand defines the running continuous control cost $r(u_x, u_y)$. Substituting $\mathbf{B}\mathbf{B}^\top = \text{diag}(2D_\parallel, 2D_\perp)$:

$$r(u_x, u_y) = \frac{1}{2} \mathbf{u}_t^\top \begin{pmatrix} \frac{1}{2D_\parallel} & 0 \\ 0 & \frac{1}{2D_\perp} \end{pmatrix} \mathbf{u}_t = \frac{u_x^2}{4D_\parallel} + \frac{u_y^2}{4D_\perp}. \quad (23)$$

Minimizing the KL divergence between path measures is therefore mathematically equivalent to minimizing an anisotropic quadratic control cost, weighted inversely by the directional noise components.

General Stochastic Hamilton–Jacobi–Bellman Equation

The general form of the continuous stochastic Hamilton–Jacobi–Bellman (HJB) equation for the value function $V(x, y)$ is written as:

$$0 = \min_{\mathbf{u}} [q(x, y) + r(\mathbf{u}) + \mathbf{u} \cdot \nabla V + \text{Itô correction}], \quad (24)$$

where:

- $q(x, y)$ is the continuous state cost we wish to recover,
- $r(\mathbf{u})$ is the running control cost,
- $\mathbf{u} \cdot \nabla V$ is the drift advection term (since passive environmental drift is zero),
- The Itô correction accounts for the second-order spatial diffusion.

Explicit Substitution into the HJB Equation

We write down the explicit forms of each term in the HJB equation for our system. The control cost derived from Girsanov’s theorem is:

$$r(u_x, u_y) = \frac{u_x^2}{4D_\parallel} + \frac{u_y^2}{4D_\perp}. \quad (25)$$

The drift advection term is:

$$\mathbf{u} \cdot \nabla V = u_x \partial_x V + u_y \partial_y V. \quad (26)$$

The Itô correction term for our diagonal anisotropic diffusion tensor is:

$$\text{Itô correction} = D_\parallel \partial_{xx} V + D_\perp \partial_{yy} V. \quad (27)$$

Substituting these explicit components into the general stochastic HJB equation yields:

$$0 = \min_{u_x, u_y} \left[q(x, y) + \frac{u_x^2}{4D_{\parallel}} + \frac{u_y^2}{4D_{\perp}} + u_x \partial_x V + u_y \partial_y V + D_{\parallel} \partial_{xx} V + D_{\perp} \partial_{yy} V \right]. \quad (28)$$

Isolating the control-dependent terms inside the minimization bracket:

$$0 = \min_{u_x, u_y} \left[\frac{u_x^2}{4D_{\parallel}} + u_x \partial_x V + \frac{u_y^2}{4D_{\perp}} + u_y \partial_y V \right] + q(x, y) + D_{\parallel} \partial_{xx} V + D_{\perp} \partial_{yy} V. \quad (29)$$

Minimization for Optimal Control

To find the optimal control policy, we minimize the control-dependent terms in the bracket:

$$\min_{u_x, u_y} \left[\frac{u_x^2}{4D_{\parallel}} + u_x \partial_x V + \frac{u_y^2}{4D_{\perp}} + u_y \partial_y V \right]. \quad (30)$$

Taking partial derivatives with respect to u_x and u_y and setting them to zero yields the first-order optimality conditions:

$$\frac{\partial}{\partial u_x} \left(\frac{u_x^2}{4D_{\parallel}} + u_x \partial_x V \right) = \frac{u_x}{2D_{\parallel}} + \partial_x V = 0, \quad (31)$$

$$\frac{\partial}{\partial u_y} \left(\frac{u_y^2}{4D_{\perp}} + u_y \partial_y V \right) = \frac{u_y}{2D_{\perp}} + \partial_y V = 0. \quad (32)$$

Solving for the optimal controls u_x^* and u_y^* :

$$u_x^* = -2D_{\parallel} \partial_x V, \quad (33)$$

$$u_y^* = -2D_{\perp} \partial_y V. \quad (34)$$

In vector form, the optimal control is $\mathbf{u}^* = -2\mathbf{D}\nabla V$, where $\mathbf{D} = \text{diag}(D_{\parallel}, D_{\perp})$. This control generates the entire macroscopic velocity field of the optimal agent.

Substitution Back into the HJB Equation

We substitute the optimal controls u_x^* and u_y^* back into the minimized bracket. For the x -component:

$$\frac{(u_x^*)^2}{4D_{\parallel}} + u_x^* \partial_x V = \frac{(-2D_{\parallel} \partial_x V)^2}{4D_{\parallel}} + (-2D_{\parallel} \partial_x V) \partial_x V = D_{\parallel} (\partial_x V)^2 - 2D_{\parallel} (\partial_x V)^2 = -D_{\parallel} (\partial_x V)^2. \quad (35)$$

Similarly, for the y -component:

$$\frac{(u_y^*)^2}{4D_\perp} + u_y^* \partial_y V = \frac{(-2D_\perp \partial_y V)^2}{4D_\perp} + (-2D_\perp \partial_y V) \partial_y V = -D_\perp (\partial_y V)^2. \quad (36)$$

Substituting these minimized terms back yields the non-linear partial differential equation governing the value function $V(x, y)$:

$$0 = q(x, y) - D_\parallel (\partial_x V)^2 - D_\perp (\partial_y V)^2 + D_\parallel \partial_{xx} V + D_\perp \partial_{yy} V. \quad (37)$$

Recovering the State Cost via Direct Substitution

Since the optimal control $\mathbf{u}^* = (u_x^*, u_y^*)$ is known from the optimal policy, we recover the state cost $q(x, y)$ directly by rearranging the minimized HJB equation:

$$q(x, y) = D_\parallel (\partial_x V)^2 + D_\perp (\partial_y V)^2 - D_\parallel \partial_{xx} V - D_\perp \partial_{yy} V. \quad (38)$$

Extracting the Value Function Gradients

From the first-order optimality conditions, the optimal control is related to the gradient of the value function by:

$$u_x^* = -2D_\parallel \partial_x V, \quad (39)$$

$$u_y^* = -2D_\perp \partial_y V. \quad (40)$$

Inverting these relations allows us to express the first spatial derivatives of the value function directly in terms of the known optimal control:

$$\partial_x V = -\frac{u_x^*}{2D_\parallel}, \quad (41)$$

$$\partial_y V = -\frac{u_y^*}{2D_\perp}. \quad (42)$$

Computing the Second Derivatives

To evaluate the Itô correction terms in the isolated $q(x, y)$ equation, we require the second spatial derivatives of the value function. Differentiating the first derivatives with respect to x and y respectively yields:

$$\partial_{xx} V = \partial_x \left(-\frac{u_x^*}{2D_\parallel} \right) = -\frac{1}{2D_\parallel} \partial_x u_x^*, \quad (43)$$

$$\partial_{yy} V = \partial_y \left(-\frac{u_y^*}{2D_\perp} \right) = -\frac{1}{2D_\perp} \partial_y u_y^*. \quad (44)$$

Closed-Form Expression for the State Cost

We now substitute the expressions for $\partial_x V$, $\partial_y V$, $\partial_{xx} V$, and $\partial_{yy} V$ back into our isolated, uncorrupted equation for $q(x, y)$:

$$q(x, y) = D_{\parallel} \left(-\frac{u_x^*}{2D_{\parallel}} \right)^2 + D_{\perp} \left(-\frac{u_y^*}{2D_{\perp}} \right)^2 - D_{\parallel} \left(-\frac{1}{2D_{\parallel}} \partial_x u_x^* \right) - D_{\perp} \left(-\frac{1}{2D_{\perp}} \partial_y u_y^* \right). \quad (45)$$

Simplifying the algebraic terms, we arrive at the exact closed-form equation for the state cost:

$$q(x, y) = \frac{(u_x^*)^2}{4D_{\parallel}} + \frac{(u_y^*)^2}{4D_{\perp}} + \frac{1}{2} \partial_x u_x^* + \frac{1}{2} \partial_y u_y^*. \quad (46)$$

This result demonstrates that the recovered state cost $q(x, y)$ consists of two primary physical contributions:

1. **Control Effort (Kinetic Energy):** $\frac{(u_x^*)^2}{4D_{\parallel}} + \frac{(u_y^*)^2}{4D_{\perp}}$, penalizing active displacement against diffusion noise.
2. **Divergence Penalty:** $\frac{1}{2}(\partial_x u_x^* + \partial_y u_y^*)$, accounting for the spatial variation and trajectory convergence of the optimal policy.

Explicit Substitution for the State Cost

Because the agent's policy generates the total velocity field, the optimal control components $\mathbf{u}^* = (u_x^*, u_y^*)$ are equal to the total drift components derived from the linearized Fokker–Planck equation:

$$u_x^* = v_0 - (1 - \gamma) \frac{\alpha \epsilon^2}{2} \partial_x O, \quad (47)$$

$$u_y^* = \kappa \theta - (1 - \gamma) \frac{\alpha \epsilon^2}{2} \partial_y O. \quad (48)$$

Computing the Divergence Terms

To evaluate the divergence term $\frac{1}{2}(\partial_x u_x^* + \partial_y u_y^*)$, we differentiate the control components with respect to space. Note that v_0 and κ are constants, and θ is the small angle to the goal.

The x -derivative is:

$$\partial_x u_x^* = -(1 - \gamma) \frac{\alpha \epsilon^2}{2} \partial_{xx} O. \quad (49)$$

The y -derivative is:

$$\partial_y u_y^* = \kappa \partial_y \theta - (1 - \gamma) \frac{\alpha \epsilon^2}{2} \partial_{yy} O. \quad (50)$$

Summing these derivatives gives the divergence of the optimal control:

$$\partial_x u_x^* + \partial_y u_y^* = \kappa \partial_y \theta - (1 - \gamma) \frac{\alpha \epsilon^2}{2} \nabla^2 O, \quad (51)$$

where $\nabla^2 O = \partial_{xx} O + \partial_{yy} O$ is the Laplacian of the obstacle density field.

Final Closed-Form State Cost

Substituting the explicit control components and the divergence sum into the state cost equation yields:

$$q(x, y) = \frac{(u_x^*)^2}{4D_{\parallel}} + \frac{(u_y^*)^2}{4D_{\perp}} + \frac{1}{2}(\partial_x u_x^* + \partial_y u_y^*). \quad (52)$$

Expanding this explicitly, we obtain the final recovered continuous state cost:

$$\begin{aligned} q(x, y) = & \frac{1}{4D_{\parallel}} \left(v_0 - (1 - \gamma) \frac{\alpha \epsilon^2}{2} \partial_x O \right)^2 \\ & + \frac{1}{4D_{\perp}} \left(\kappa \theta - (1 - \gamma) \frac{\alpha \epsilon^2}{2} \partial_y O \right)^2 \\ & + \frac{\kappa}{2} \partial_y \theta - \frac{(1 - \gamma) \alpha \epsilon^2}{4} \nabla^2 O. \end{aligned} \quad (53)$$

This final expression reveals the physical structure of the recovered state cost $q(x, y)$:

1. **Control Effort (Kinetic Energy):** The first two terms represent the squared magnitude of the optimal control, weighted inversely by the anisotropic diffusion noise $4D_{\parallel}$ and $4D_{\perp}$. This penalizes rapid accelerations or moving against repulsive obstacle gradients.
2. **Transverse Restoring Potential:** The term $\frac{\kappa}{2} \partial_y \theta$ arises from the geocentric policy's angular dependence, penalizing paths where the goal angle changes rapidly in the transverse direction.
3. **Obstacle Curvature Repulsion:** The term $-\frac{(1-\gamma)\alpha\epsilon^2}{4} \nabla^2 O$ penalizes the agent based on the Laplacian (curvature) of the obstacle field, pushing the agent away from concave traps or tight corridors where obstacle density is concentrated.